

PLAYDB – Video Game Database & Marketplace System

DELIVERABLE 1(B)

Scrum Methodology Document

Course: Software Engineering

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Submission Date: 28th February 2026

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1. Development Methodology – Scrum

The PlayDB project will be developed using the Scrum agile methodology. Scrum is an iterative and incremental framework that allows the team to develop the system in small, manageable phases called sprints. This approach enables continuous improvement, flexibility, and regular evaluation of progress.

Each sprint will have a fixed duration of two weeks. At the beginning of every sprint, the team will conduct a sprint planning meeting to select user stories from the product backlog and define sprint goals. Tasks will be distributed among team members based on their assigned roles and expertise.

During the sprint, progress will be tracked using a Scrum board (Trello), where tasks move through stages such as To Do, In Progress, and Done. Regular meetings will be held to discuss progress, challenges, and next steps.

At the end of each sprint, the team will review completed work, update documentation, and prepare for the next sprint. This iterative workflow ensures continuous development, testing, and improvement of the PlayDB system.

2. Sprint 1 Backlog

User Management Module

The User Management Module has been selected for Sprint 1 because it forms the foundation of the PlayDB system. All other modules such as game tracking, marketplace functionality, and AI recommendations depend on user authentication and profile management.

This module will focus on implementing basic user-related functionalities including registration, login, logout, and profile management. Completing this module in Sprint 1 will ensure that a secure and structured user system is established before integrating advanced features in later sprints.

3. User Stories

User Story 1 – User Registration

As a new user

I want to create an account

So that I can access the PlayDB system and track my gaming activity.

Sub-Tasks:

- Design registration form (UI)
- Validate user inputs (email, password, username)
- Store user data in database
- Display success or error message

User Story 2 – User Login

As a registered user

I want to log in to my account

So that I can access my personalized dashboard.

Sub-Tasks:

- Design login form
- Validate credentials against database
- Create user session
- Redirect to dashboard on success
- Display error message on invalid login

User Story 3 – View Profile

As a registered user

I want to view my profile information

So that I can see my account details.

Sub-Tasks:

- Create profile page UI
- Retrieve user data from database
- Display username, email, and basic information

4. Structured Specifications

Structured Specification – User Registration

1. Input:

- Username
- Email address
- Password

2. Processing:

- Validate that all fields are filled.
- Check if email format is valid.
- Verify password meets minimum security requirements.
- Check database to ensure user does not already exist.
- Encrypt password.
- Store user data in the database.

3. Output:

- Success message: “Registration successful.” OR
- Error message indicating invalid input or existing user.

4. Preconditions:

- User is not already registered.
- Database connection is active.

5. Postconditions:

- New user account is successfully created and stored in the system.

Structured Specification – User Login

1. Input:

- Email/Username
- Password

2. Processing:

- Validate that fields are not empty.
- Retrieve user record from database.
- Compare entered password with encrypted password.
- If valid, create user session.

3. Output:

- Redirect to user dashboard on successful login.
OR
- Error message: “Invalid credentials.”

4. Preconditions:

- User account already exists.
- Database is accessible.

5. Postconditions:

- User session is created and user gains access to the system.

Structured Specification – View Profile

1. Input:

- Profile page request from logged-in user.

2. Processing:

- Retrieve user data from database.
- Fetch username, email, and other stored information.
- Display data on profile page.

3. Output:

- User profile information displayed on screen.

4. Preconditions:

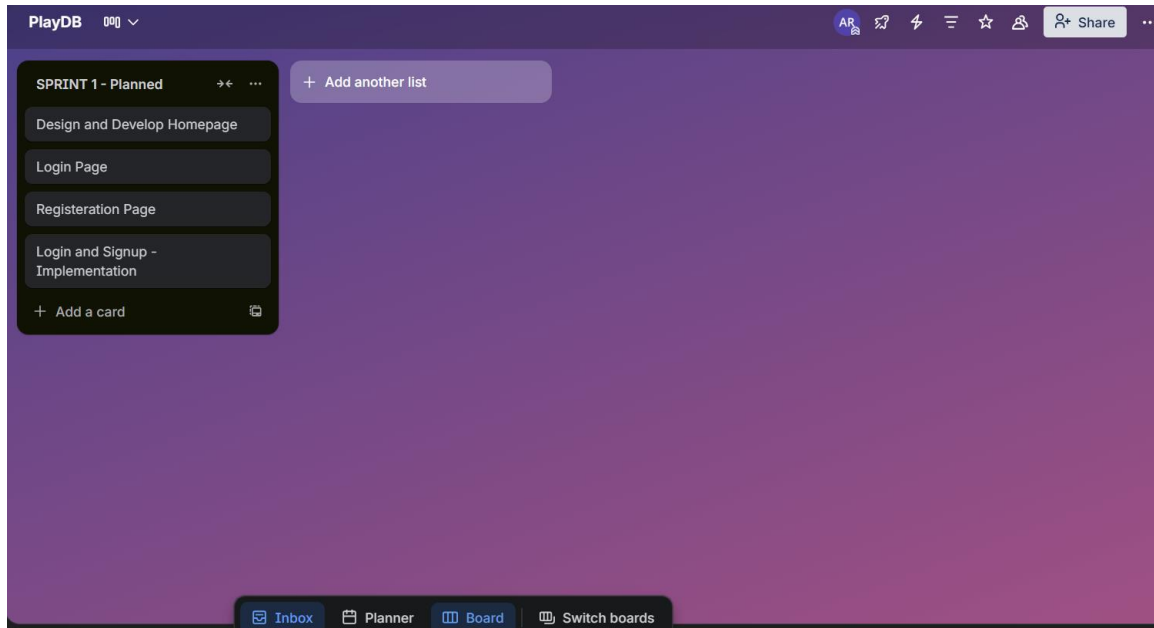
- User must be logged in.
- User data must exist in database.

5. Postconditions:

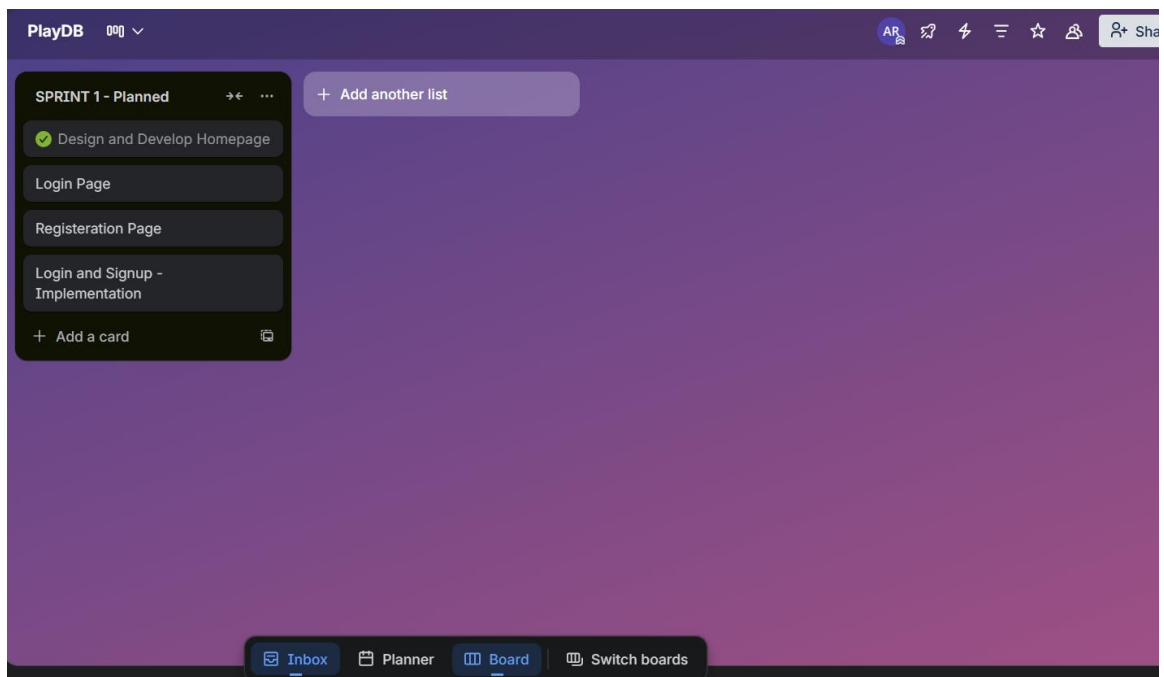
- Profile information is visible to the user.

5. Scrum Board Snapshots

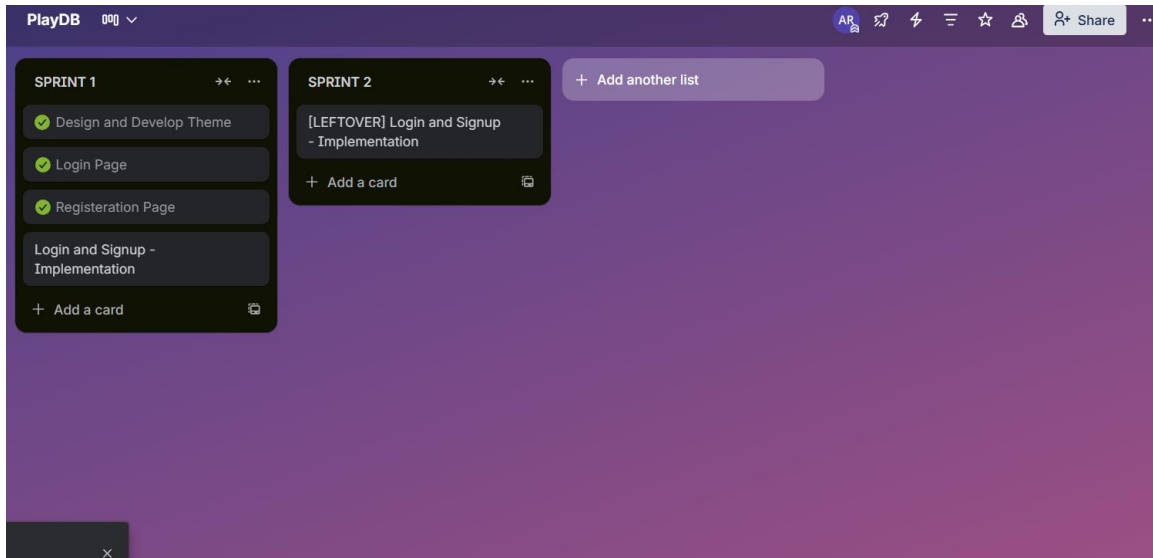
Initial Backlog



Mid Sprint



End Sprint



6. Product Burndown Chart

